

Sui Skills MCP

The Sui Stack as one-job-per-tool primitives, over the Model Context Protocol.

Capability Reference · 21 tools · Generated 2026-05-29

This MCP server is the product: plug it into Claude Desktop, Cursor, Windsurf, or any MCP-aware agent and you get one-line tools for reading chain state, building any Sui transaction, simulating it, submitting it, and storing data on Walrus. **Suisai** — the Sui teaching agent — is a showcase of it: a working app built entirely on these tools, demonstrating what an agent can do once it speaks Sui.

1. What it is

Most Sui SDKs assume you are writing *application* code. This package assumes you are writing *agent* code: short tool calls, structured JSON in, structured JSON out, no React, no UI. The agent decides what to do; the tool does exactly one thing and returns machine-readable output.

The server speaks JSON-RPC 2.0 over **stdio** per the MCP specification. There is no network listener — the host agent process spawns the binary, talks to it over stdin/stdout, and supervises its lifecycle.

2. Design principles

Principle	What it means in practice
No private keys	Tools that produce transactions return base64 transaction bytes. The host signs and submits. A tool that holds a key is a tool that can spend money — so this toolkit never holds one.
One job per tool	No mega-tools with twelve flags. Each tool is unambiguous, which sharpens the calling agent's reasoning about when to use it.
Structured output	Every successful return is a JSON string in the text content block. Agents parse structured data far more reliably than prose.

3. The build loop

Write operations follow a four-step, non-custodial loop. The toolkit covers steps 1, 2 and 4; the host owns the key and performs step 3.

1. `*_build` tool → returns base64 unsigned tx bytes
2. `sui_dry_run` → `simulate`: status + gas, no spend
3. host signs the bytes (outside the toolkit — the host holds the key)
4. `sui_execute_signed_tx` → submit signed bytes, return digest + effects

Security invariant: every transaction-building tool ends by returning bytes, never by signing. The private key never enters the MCP process.

4. Install & connect

```
npm install -g @suisai/sui-skills-mcp
```

Then register it with an MCP host. For Claude Desktop, edit `claude_desktop_config.json`:

```
{
  "mcpServers": {
    "sui": {
      "command": "npX",
      "args": ["-y", "@suisai/sui-skills-mcp"]
    }
  }
}
```

Restart the host. All tools are exposed under the sui server namespace.

5. Tool reference

Twenty-one tools across four groups. Every tool accepts a network argument (testnet · mainnet · devnet, default testnet) except the two Walrus tools, which target Walrus endpoints directly.

5.1 Read the chain

Tool	What it does
sui_resolve_address	SuiNS name → 0x address (idempotent on 0x input)
sui_get_balance	SUI balance in MIST + human-readable SUI
sui_get_all_balances	Every coin balance a wallet holds, not just SUI
sui_get_object	Any object's type, owner, version, fields, and Display
sui_get_owned_objects	List objects by owner, filter by struct type, paginated
sui_get_owned_badges	List a wallet's Suisai quest-completion badges
sui_get_coins	List coin objects of a type (the ids you spend), paginated
sui_get_transaction	Look up a finalized tx by digest: status, gas, changes
sui_get_reference_gas_price	Current network reference gas price (MIST)
sui_get_dynamic_fields	List an object's dynamic fields (Tables, Bags)
sui_deepbook_quote	Read-only DeepBook v3 quote: expected out + DEEP fee

sui_resolve_address

If given a 0x address, returns it canonicalized (lowercased). Otherwise treats the input as a SuiNS name and resolves it. Use this whenever a user gives a human-readable name.

Input	Type	Req	Notes
name_or_address	string	yes	A SuiNS name or 0x address
network	enum	no	default testnet

```
→ { "address": "0x...", "source": "suins", "name": "alice.sui" }
```

sui_get_balance / sui_get_all_balances

sui_get_balance returns the native SUI balance for one address; sui_get_all_balances returns every coin type held. Both take address + network.

```
→ { "coin_type": "0x2::sui::SUI", "total_mist": "1500000000",
    "total_sui": 1.5, "coin_object_count": 3 }
```

sui_get_object

The general read primitive — every other read tool is a special case of it. Returns Move type, owner, version, digest, content fields, and Display metadata for any object id.

```
→ { "type": "0x...:badge::Badge", "owner": {...}, "version": "...",  
    "fields": {...}, "display": {...} }
```

sui_get_owned_objects

Lists objects owned by an address, optionally filtered to a single fully-qualified Move struct type. Paginated — pass the returned `next_cursor` to continue.

Input	Type	Req	Notes
address	string	yes	Wallet to list
struct_type	string	no	e.g. <code>0x2::coin::Coin<0x2::sui::SUI></code>
cursor	string	no	Pagination cursor
limit	number	no	1–50, default 50
network	enum	no	default testnet

sui_get_owned_badges

Lists SuiSei badges (soulbound quest-completion NFTs) owned by an address — useful for showing curriculum progress. Resolves the badge package from the `badge_package` argument, the `SUISEI_BADGE_PACKAGE` env var, or a built-in per-network default.

```
→ { "count": 3, "badges": [ { "object_id": "...",  
    "quest_id": "zklogin", "quest_number": 1, "minted_at_ms": "..." } ] }
```

sui_get_coins

Unlike `sui_get_balance` (which sums), this returns the concrete `coin_object_id` values an agent needs to spend or split in a transaction. Defaults to native SUI; pass `coin_type` for any other coin. Paginated.

```
→ { "coin_type": "0x2::sui::SUI", "count": 4, "has_next_page": false,  
    "coins": [ { "coin_object_id": "...", "balance": "10000000000" }, ... ] }
```

sui_get_transaction

Fetch a finalized transaction by digest — the same digest `sui_execute_signed_tx` returns. Returns status, gas used, balance changes, timestamp, and event count.

```
→ { "digest": "...", "status": "success", "timestamp_ms": "...",  
    "gas_used": {...}, "balance_changes": [...], "events_count": 1 }
```

sui_get_reference_gas_price / sui_get_dynamic_fields

`sui_get_reference_gas_price` returns the current network gas price in MIST (takes only network).

`sui_get_dynamic_fields` lists the dynamic fields attached to a parent object — how Sui stores Tables, Bags, and other on-chain collections — returning each field's name, type, and child object id (paginated).

5.2 Build a transaction (returns unsigned bytes)

Tool	What it does
<code>sui_move_call</code>	Build a call to ANY Move entry function — the universal write
<code>sui_transfer</code>	Build a transfer of SUI and/or whole objects
<code>sui_stake</code>	Build a native staking delegation to a validator
<code>sui_unstake</code>	Build a withdraw-stake for a <code>StakedSui</code> object
<code>sui_mint_badge</code>	Build a SuiSei completion-badge mint
<code>sui_deepbook_swap</code>	Build a DeepBook v3 market swap (CLOB liquidity)

sui_move_call — the universal write primitive

Builds an unsigned transaction calling any Move entry function. Arguments are encoded as strings so the schema stays flat and unambiguous:

- `object:<id>` — an owned or shared object by id
- `pure:<type>:<value>` — type is address, id, bool, string, or u8/u16/u32/u64/u128/u256

Vector and nested-struct arguments are intentionally out of scope for the generic tool. Use a purpose-built tool (e.g. `sui_transfer`) for those, so the generic schema stays unambiguous for the calling agent.

Input	Type	Req	Notes
target	string	yes	0xpkg::module::function
type_arguments	string[]	no	generics, e.g. ["0x2::sui::SUI"]
arguments	string[]	no	encoded as above
sender	string	yes	0x address that will sign & pay
network	enum	no	default testnet

```
→ { "tx_bytes_base64": "...", "target": "0x...::m::f",  
    "sender": "0x...", "next_step": "Dry-run, then sign and submit." }
```

sui_transfer

Provide `amount_mist` to send SUI (split from the gas coin) and/or `object_ids` to send whole objects. At least one is required.

Input	Type	Req	Notes
sender	string	yes	Sends & pays
recipient	string	yes	Receives
amount_mist	string	no*	SUI in MIST (string, avoids precision loss)
object_ids	string[]	no*	Whole objects to transfer

* one of `amount_mist` or `object_ids` is required.

sui_stake / sui_unstake

`sui_stake` splits `amount_mist` from gas and delegates to a validator via `0x3::sui_system::request_add_stake`; the `StakedSui` object lands in the sender once signed. `sui_unstake` withdraws a `StakedSui` by id via `request_withdraw_stake`, returning principal plus rewards.

Tool	Required inputs
<code>sui_stake</code>	sender, amount_mist, validator
<code>sui_unstake</code>	sender, staked_sui_id

sui_mint_badge

Builds a PTB that mints a Suisei completion badge to a recipient through the canonical badge module. Returns unsigned bytes; the caller signs and submits.

Input	Type	Req	Notes
recipient	string	yes	0x address receiving the badge
quest_id	string	yes	e.g. zklogin, sponsored
quest_number	number	yes	1–255
badge_package	string	yes	Move package with the badge module

sui_deepbook_quote

Read-only DeepBook v3 quote via devInspect — no gas, no signing, no funds. Given an input amount and direction, returns the expected output and the DEEP fee required, so an agent can size min_out before building a swap. Same pool selection as the swap (known key, or explicit pool_id + base_type + quote_type). All values are raw smallest-units, so expected_out feeds straight into the swap's min_out.

```
→ { "expected_out": "900000", "deep_required": "27770",  
    "base_out": "0", "quote_out": "900000", "direction": "base_to_quote" }
```

sui_deepbook_swap

Builds the exact market-swap PTB the official DeepBook v3 SDK emits (pool::swap_exact_base_for_quote / swap_exact_quote_for_base), on the current @mysten/sui v1, so the toolkit stays non-custodial. Pass a known pool key (SUI_DBUSDC on testnet, SUI_USDC on mainnet) or explicit pool_id + base_type + quote_type. amount and min_out are raw smallest-unit strings (caller handles decimals and slippage). DeepBook charges fees in DEEP: whitelisted pools take deep_amount 0, others require the sender to hold DEEP. The three output coins are transferred back to the sender.

Input	Type	Req	Notes
sender	string	yes	Signs, pays, receives output
direction	enum	yes	base_to_quote quote_to_base
amount	string	yes	Input in smallest units
min_out	string	no	Slippage floor (default 0)
deep_amount	string	no	DEEP fee (default 0; whitelisted pools)
pool	string	no*	Known pool key, e.g. SUI_DBUSDC
pool_id / base_type / quote_type	string	no*	Explicit pool (overrides key)

* provide a known pool key or the explicit pool_id + base_type + quote_type trio.

5.3 Simulate & submit

Tool	What it does
sui_dry_run	Simulate built tx bytes (status + gas, no spend)
sui_execute_signed_tx	Submit host-signed bytes, return digest + effects

sui_dry_run

Simulates an unsigned transaction without spending gas, so an agent can verify a builder result before asking the host to sign. Takes tx_bytes_base64.

```
→ { "status": "success", "gas_used": {...}, "balance_changes": [...],  
    "object_changes_count": 2, "events_count": 0 }
```

sui_execute_signed_tx

Submits a host-signed transaction. Takes the same tx_bytes_base64 that were signed plus one or more base64 signatures. The toolkit holds no keys — signing happens in the host.

```
→ { "digest": "...", "status": "success", "gas_used": {...},  
    "balance_changes": [...], "events_count": 1 }
```

5.4 Walrus storage

Tool	Key inputs	Returns
walrus_publish	content, encoding (utf8/base64), epochs (default 5), publisher_url?	blob_id, status, size_bytes
walrus_fetch	blob_id, as (utf8/base64), aggregator_url?	content, encoding, size_bytes

6. Configuration

Variable	Purpose
SUISEI_BADGE_PACKAGE	Default badge package id for sui_get_owned_badges when not passed explicitly.
Walrus endpoints	Default to the public Walrus testnet publisher/aggregator; override per-call via publisher_url / aggregator_url.
Network	Defaults to testnet; pass network on any tool to switch.

7. Security model

- **Non-custodial by construction.** No tool accepts, stores, or derives a private key. Builder tools return unsigned bytes; only sui_execute_signed_tx touches a signature, and that signature is produced by the host.
- **No ambient network surface.** Stdio transport only — the server opens no port and accepts no inbound connections.
- **Verify before spend.** The documented loop routes every write through sui_dry_run before signing, so an agent can confirm gas and effects up front.

© 2026 Suiwei · @suiwei/sui-skills-mcp v0.1.0 · MIT License. The build loop is: a builder tool returns base64 tx bytes → dry-run to verify → the host signs → execute to submit. The toolkit never holds keys.